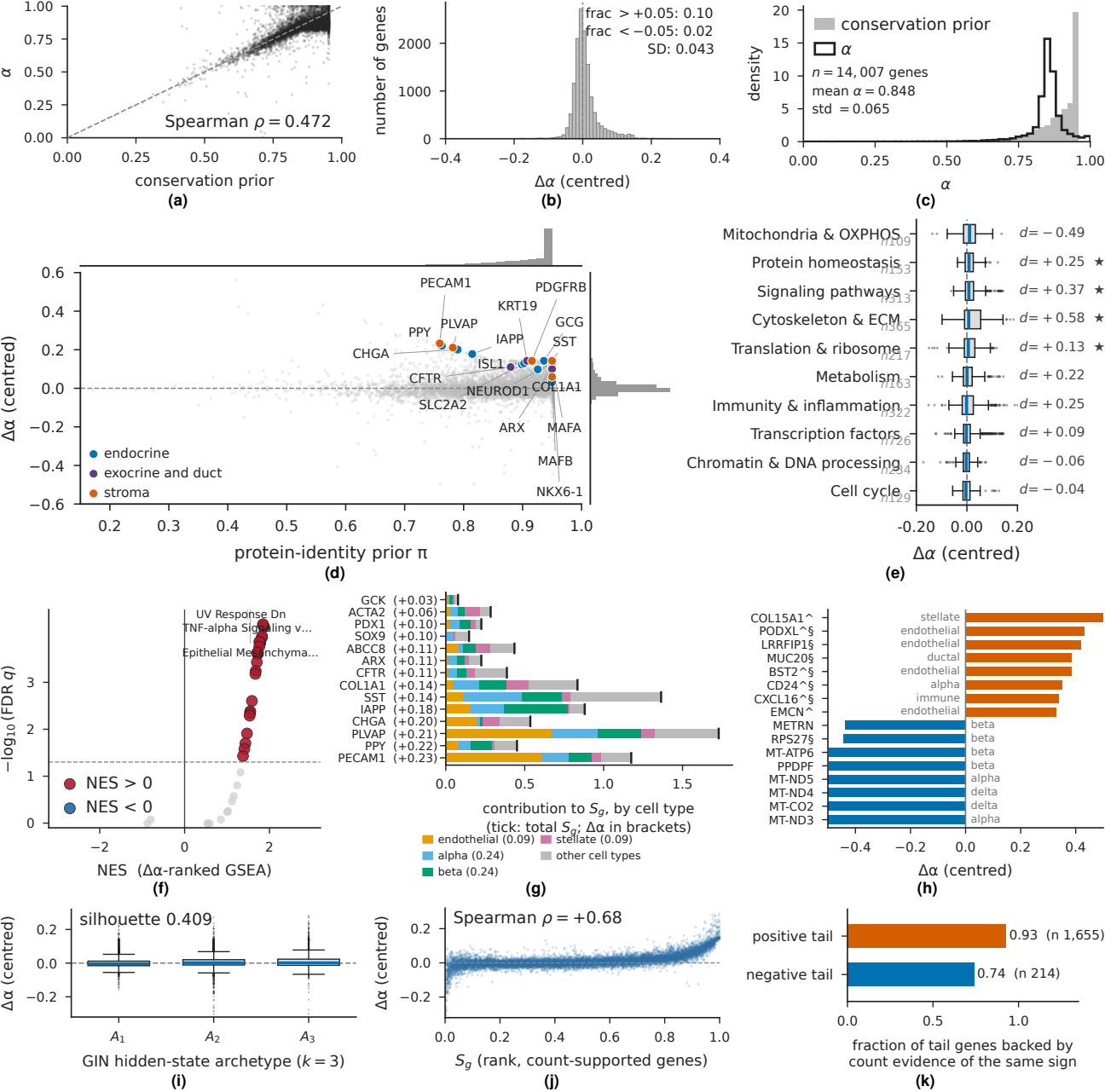

Spearman ρ = 0.472
conservation prior
α
(a)
frac > +0.05: 0.10
frac < −0.05: 0.02
SD: 0.043
number of genes
Δα (centred)
(b)
conservation prior
α
n = 14,007 genes
mean α = 0.848
std = 0.065
density
α
(c)
PECAM1
PDGFRB
KRT19
PLVAP
GCG
PPY
IAPP
SST
CHGA
ISL1
CFTR
NEUROD1
COL1A1
SLC2A2
ARX
MAFA
MAFB
NKX6-1
endocrine
exocrine and duct
stroma
Δα (centred)
protein-identity prior π
(d)
Mitochondria & OXPHOS n109 d = −0.49
Protein homeostasis n153 d = +0.25 ★
Signaling pathways n313 d = +0.37 ★
Cytoskeleton & ECM n365 d = +0.58 ★
Translation & ribosome n217 d = +0.13 ★
Metabolism n163 d = +0.22
Immunity & inflammation n322 d = +0.25
Transcription factors n726 d = +0.09
Chromatin & DNA processing n294 d = −0.06
Cell cycle n129 d = −0.04
Δα (centred)
(e)
UV Response Dn
TNF-alpha Signaling v...
Epithelial Mesenchyma...
NES > 0
NES < 0
−log10 (FDR q)
NES (Δα-ranked GSEA)
(f)
GCK (+0.03)
ACTA2 (+0.06)
PDX1 (+0.10)
SOX9 (+0.10)
ABCC8 (+0.11)
ARX (+0.11)
CFTR (+0.11)
COL1A1 (+0.14)
SST (+0.14)
IAPP (+0.18)
CHGA (+0.20)
PLVAP (+0.21)
PPY (+0.22)
PECAM1 (+0.23)
contribution to $S_g$, by cell type
(tick: total $S_g$; Δα in brackets)
endothelial (0.09)
stellate (0.09)
alpha (0.24)
other cell types
beta (0.24)
(g)
COL15A1^ stellate
PODXL^§ endothelial
LRRFIP1§ endothelial
MUC20§ ductal
BST2^§ endothelial
CD24^§ alpha
CXCL16^§ immune
EMCN^ endothelial
METRN beta
RPS27§ beta
MT-ATP6 beta
PPDPF beta
MT-ND5 alpha
MT-ND4 delta
MT-CO2 delta
MT-ND3 alpha
Δα (centred)
(h)
silhouette 0.409
Δα (centred)
$A_1$
$A_2$
$A_3$
GIN hidden-state archetype ($k = 3$)
(i)
Spearman ρ = +0.68
Δα (centred)
$S_g$ (rank, count-supported genes)
(j)
positive tail 0.93 (n 1,655)
negative tail 0.74 (n 214)
fraction of tail genes backed by
count evidence of the same sign
(k)